

Sohhom Bandyopadhyay

Quantitative Research & Applied ML · PhD, Cognitive Science

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SUMMARY

Quantitative researcher and applied ML engineer with a PhD in Cognitive Science (IIT Gandhinagar) and 4+ years of industry experience across LLMs, classical ML, and signal processing. Recent work is in markets — trading copilots, options analytics, and an end-to-end research system for Indian equities built solo. My background is in measurement under uncertainty: designing the experiment, choosing what to measure, and establishing whether an effect is real or an artifact of the method. That instinct transfers directly to backtesting, where the default outcome is fooling yourself. I am equally willing to report that something works and that something does not.

EXPERIENCE

Independent Researcher — Quantitative Trading

Self-directed · Kolkata, India · *July 2024 – Present*

- Building an end-to-end research and trading system for Indian equities (NSE/BSE): tick and OHLCV ingestion, feature and label engineering, signal generation, walk-forward validation, and execution logic.
- Wrote reusable components for transaction-cost modelling, strategy evaluation, and risk metrics; every candidate strategy passes a cost-and-survivorship break-even test before anything else is measured, which kills most ideas early and cheaply.
- Treat regime awareness and selectivity as first-class: deciding *when not to trade* is part of the strategy rather than a filter bolted on afterwards.
- Emphasis on falsification — look-ahead audits, label/timestamp alignment checks, and per-regime PnL attribution to establish that a result is not carried by a handful of days.
- Stack: Python, Polars, pandas, NumPy, SciPy, scikit-learn, broker APIs.

Applied AI Engineer — Trading Systems

Punch (Market Pulse Securities) · Remote · *October 2025 – May 2026*

- Built LLM-powered trading copilots that read live chart context and answered interactive market questions, architected around **symbolic reduction** — numeric series and detected geometry reduced to discrete semantic state before reaching the model, so every quantitative claim traced to deterministic code rather than model arithmetic.
- Shipped rule-based technical-pattern recognition (head-and-shoulders, double tops/bottoms, cup-and-handle, wedges, triangles, flags, pennants, trendline and support/resistance breaks) over OHLC data, surfaced as live chart annotations; ported the geometry to JavaScript for client-side execution with numerical parity across runtimes.
- Designed a **contract-driven agentic pipeline** for pattern detection: separate written contracts for specification, detection, rendering and evaluation, with a vision-model rubric scoring rendered charts against the spec's own numbered rules — making spec-versus-implementation disagreement measurable and localisable to a named rule.
- Delivered derivatives (F&O) analytics dashboards covering open-interest dynamics, OI and volume profiles, gamma exposure (GEX), max pain, and implied-volatility percentile conditions across expiries and multiple timeframes.
- Developed strategy-selection logic for ~20 multi-leg options structures, conditioned on a multi-axis market-regime classification (direction, volatility level, volatility premium, term structure, skew), with independent gating for regime fit, structural validity and time-to-expiry.
- Added **portfolio-level risk management** — netting-aware aggregate Greeks, per-trade gamma and vega budgets driving position sizing, and exchange SPAN margin modelling — plus a backtesting and paper-trading harness with an exact transaction-cost model, and a line-by-line audit of all 20 strategy constructions against standard options references.
- Sole ML contributor on the engagement; worked directly with product and engineering leads, including pushing back on approaches that would not have survived contact with real data.

Artificial Intelligence Researcher

Playpower Labs · Remote (India) · *September 2023 – June 2024*

- Senior individual contributor reporting to the CEO; embedded with product and engineering as needed.
- Led research on **human preference elicitation for AI alignment** — the question of whether the interface used to collect preference data systematically changes the preferences recovered. Surveyed and instrumented rating scales, pairwise and n-alternative forced choice, like/dislike, best-worst scaling, categorical binning and point allocation, synthesising literature across psychometrics, experimental economics, computational social choice and RLHF.
- Built the instrumentation to test it: a custom n-alternative forced-choice plugin (2-, 4- and 6-choice image selection) for a browser psychophysics framework that had no such capability, plus deployed collection apps with cross-session participant tracking and a live results dashboard.
- Ran crowdsourced studies end to end — stimulus generation, deployment, participant-quality screening (response-time outlier detection to platform criteria; per-participant agreement-with-mean correlation), and analysis.

- Diagnosed a **disconnected comparison graph** in an existing preference dataset — comparisons never crossed category boundaries, leaving Bradley–Terry and Elo scores unidentifiable across clusters — and designed the task revision that restored connectivity.
- Re-derived a three-way within-subjects ANOVA in R from raw data with effect sizes and corrected post-hoc tests, resolving a design-specification disagreement and correcting a stale factual claim in the manuscript; contributed to a co-authored paper in *Advanced Design Research* (2024).
- Correlated a published image-preference reward model against human ratings on the same stimuli (Spearman 0.30), quantifying the gap between general “preference” and a specific judgement axis.
- Archived ~14.5k generations from a family of commercial language models shortly before deprecation, under a factorial prompt design, to preserve reproducible behaviour of models that would otherwise be unrecoverable.
- Scoped and evaluated LLM and diffusion-model applications for EdTech, including automated essay scoring; produced feasibility analyses that informed build/buy/defer decisions.

R&D Researcher — Human–AI Interaction

Playpower Labs · Remote (India) · July 2022 – April 2023

- Built and evaluated prototypes on GPT-3, ChatGPT and GPT-4 as they were released, assessing academic and commercial implications for the company’s product line — including deployed web applications with API integration, content moderation, access control and prompt/response logging.
- Fine-tuned GPT-3 base models and established the practical limits of doing so, including that the instruction-tuned (RLHF) variants were not fine-tunable at the time — a constraint that materially changed the project’s approach.
- Trained and evaluated early Stable Diffusion and diffusion-model pipelines; ran a literature review on human-preference elicitation used internally to frame product direction.
- Designed and deployed the first round of browser-based preference experiments on paid crowdsourcing platforms.

Graduate Teaching Fellow

IIT Gandhinagar · January – April 2019

- Co-taught the MSc Cognitive Science core course *Computation and Cognition* (CG 501) with Prof. Krishna Prasad Miyapuram. Independently ran 17 classes, designed quizzes, a project and the end-semester exam, and built fill-in-the-blank Jupyter assignments covering perceptron/MLP, linear regression, k-NN, and game theory.

EDUCATION

PhD, Cognitive Science

IIT Gandhinagar · 2017 – 2022 · CPI 8.68

Thesis: *Psychoacoustic assessment of active-noise-control (ANC) headphones*. The field measures how much noise ANC removes; almost nobody measures what it does to the signal you actually want to hear. Developed black-box protocols to answer the second question across four commercial headphones — a modulation transfer function for a headphone, a computational auditory-nerve model repurposed as a measuring instrument, an 80-listener speech-intelligibility experiment, and ten instrumental speech-quality metrics as a cross-check. The headline effect was small and reported as such. An external examiner described the methodological rigour as “excellent.”

MSc, Cognitive Science

IIT Gandhinagar · 2015 – 2017 · CPI 8.85

Thesis: *Functional Link Artificial Neural Network for keystroke-dynamics authentication*. Implemented a FLANN classifier from scratch in NumPy and benchmarked it against MLP, random forest, k-NN and decision trees; it gave the best AUROC and equal-error rate on free-text keystroke data.

BSc (Hons), Computer Science

Scottish Church College, University of Calcutta · 2012 – 2015 · 73.7% (*First Class*)

PUBLICATIONS

- Lomas, J. D., van der Maden, W., **Bandyopadhyay, S.**, et al. (2024). Evaluating the alignment of AI with human emotions. *Advanced Design Research*, 2(2), 88–97.
- Mukhopadhyay, D., **Bandyopadhyay, S.**, Venkatesh, E., & Menon, S. (2024). Complexity-entropy, eye-tracking and emotional responses to 40 paintings of Raja Ravi Varma. *International Journal of Arts and Technology*, 15(2), 170–207.
- Fonseca, A. F., **Bandyopadhyay, S.**, Louçã, J., & Manjaly, J. A. (2019). Caste in the news: a computational analysis of Indian newspapers. *Social Media + Society*, 5(4).

SELECTED PROJECTS

Quantitative investigations of NSE/BSE markets

Ongoing, 2024–

Rolling investigations into Indian equity and derivatives markets: transaction-cost and break-even analysis, intraday microstructure, factor decomposition, and options-volatility studies, supported by custom data pipelines and exploratory dashboards. Write-ups at sohhom.github.io/project.

Pattern reuse, from file systems to neural networks

Independent, 2026

Tested a 2018 note claiming “all data is pattern” against raw binary with no format assumptions: n-gram frequency analysis, cross-file pattern overlap, dictionary coverage curves, and pointer-overhead compression modelling. The interesting result was on the bit-alignment axis — byte boundaries turn out to be an accident of addressing rather than a property of the data.

Human preference elicitation toolbox

2023–2024

Instrumentation for comparing methods of collecting human preference data: a custom n-alternative forced-choice plugin for browser-based psychophysics, crowdsourced deployment with participant-quality screening, and a ranking-model analysis pipeline (Bradley–Terry, Elo, preference ratio) with comparison-graph connectivity diagnostics. Built to test whether the elicitation interface changes the preferences recovered — a question the alignment literature largely assumes away. Write-ups at sohohm.github.io/project.

Keystroke-based user authentication

MSc thesis, 2016–2017

Neural network built from scratch in NumPy to identify users from typing patterns (dwell and flight times, digraph features), with a proof-of-concept Flask login portal. Explored what keystroke dynamics reveal about the chunking of motor sequences.

TECHNICAL SKILLS

Languages: Python, R, MATLAB, JavaScript, SQL, C++, Bash

Quant / markets: backtesting and walk-forward validation, transaction-cost modelling, market microstructure, options Greeks and max pain, technical analysis, risk metrics, position sizing

ML / data: scikit-learn, NumPy, pandas, Polars, SciPy, statsmodels, matplotlib; statistical modelling and experiment design in R

LLMs / GenAI: prompt design, tool use and agentic flows, evaluation harnesses, Hugging Face Transformers, CLIP, Stable Diffusion / diffusion models, RLHF and preference-optimisation methods (DPO, KTO, IPO)

Human data / experiment design: preference elicitation (rating scales, n-AFC, best-worst scaling, like/dislike), Bradley–Terry and Elo ranking models, within-subjects factorial designs, inter-rater agreement, crowdsourced study deployment and participant-quality screening, jsPsych

Signal processing: scipy.signal, MATLAB, psychoacoustic and auditory-model toolboxes

Web / deployment: Flask, Node.js, MongoDB, Netlify, Vercel, Git

Domains: quantitative finance, psychoacoustics, applied ML, HCI, EdTech